Group A

1. Determination of viscosity of lubrication oil by Redwood viscometer no. 1 (At five different temperatures).
2. Determination of percentage of moisture in a coal sample
3. Determination of flash point of a given oil by Abel's apparatus
4. Determination of flash point of a given oil by Pensky Martin's apparatus
5. Determination of total solids in a water sample
6. Determination of aniline point of a given oil sample
7. Determination of Drop point of a semi-solid lubricant
8. Determination of acid value of an oil sample
9. Determination of viscosity of lubrication oil by Redwood viscometer no. 2. (at five different temperatures)
10. Determination of Steam emulsification number (SEN) of a given lubricating oil sample
11. To study the Chemical oscillations (Iodine clock reaction)
12. Potentiometric estimation of Ferrous Ammonium Sulphate using standard Potassium Dichromate Solution
13. Determination of the partition coefficient of a substance between two immiscible liquids.

14 Synthesis of polymer and nanomaterials

15 Verification of Beer- Lambert's law by visible spectroscopy

Group B

1 Determination of hardness of water by EDTA method

2 Determination of carbonates, bica
rbonates and total alkalinity of a water sample

3 Determination of chloride content of water

4 Determination of percentage purity of iron alloy by internal indicator method

5 Determination of percentage purity of iron alloy by external indicator method

Course Outcomes: This course will illustrate the principles of chemistry relevant to the study of science and engineering.The students will learn to Properties,prectical applications, analysis and determination of engineering materials like

- Lubricants (liquids and semisolid)
- Fuel oil & Coal samples,
- Water in terms of hardness, chloride content and alkalinity
- Polymers
- Application of spectroscopic techniques for

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Exercise No. 1

Aim: To determine the viscosity of given sample of lubricating oil with Redwood viscometer No.1 and to study the viscosity at various temperatures.

Articles required: Redwood viscometer No.1, Thermometers, Oil sample, 50ml Measuring cylinder, etc.

Significance of determination: Viscosity is the property of a homogeneous liquid which causes it to offer frictional resistance to its motion. Viscosity of lubricating oil tells about its suitability for lubricating purpose.
The function of a lubricant is to reduce the friction between the moving parts and to avoid the direct metal to metal contact. The moving parts remain apart due to extremely thin liquid film in between. The frictional resistance therefore, is entirely on account of the shearing of liquid which is in between the metallic surfaces. The viscosity of fluid measures the amount of internal resistance of lubricating oil.
The object of this test is to ascertain, whether the lubricating oil is sufficiently viscous to adhere to the bearings to which it is applied or is so thin that it may be squeezed out of the bearing due to high pressure & temperature.

Theory: Viscosity is that property of a fluid which is due to the resistance to its flow. It is expressed in 3 ways.
(a) Absolute dynamic viscosity expressed in C.G.S. unit as poise.
(b) Absolute kinematic viscosity expressed in C.G.S. unit as stoke.
(c) Time taken by the determining instrument in seconds noted by the name of
Instrument (Redwood viscosity).

(a)Absolute dynamic viscosity: It is the tangential force on unit area of either of two parallel layers at unit distance apart when the space between the planes is filled with the fluid and one of the planes moves relatively to the other with unit velocity in its own plane.
It is defined as the ratio of the shearing stress to the rate of shearing strain. If force F is required to keep moving an article of surface area A, in contact with the fluid separated from a stationary surface by thickness d with velocity v then,

Shearing stress $= \frac{F}{A}$

Rate of shearing strain $= \frac{v}{d}$

Absolute viscosity $= \frac{Fd}{Av}$

When u is in poise then F = 1 dyne, A = 1 sq.cm. d = 1 cm. v = 1 cm. per second.

(b) Absolute kinematic viscosity: It is obtained by dividing Dynamic viscosity by density. The metric units are stoke and centistokes. Thus if 'p' is density, then kinematic viscosity $= \frac{u}{p}$stoke.

(c) Redwood Viscosity : The timing or the rate of flow of given quantity of liquid from a given height through a calibrated capillary tube under its own weight may be used as a means for measuring viscosity and called after the name of the apparatus used for its

determination. The time in seconds required for 50 ml. of oil to gravitate through the Redwood viscometer at a given temperature is expressed as its viscosity in Redwood seconds at that temperature.
Absolute dynamic and kinematic viscosities can also be determined from the Redwood values by means of the following formula.

Absolute dynamic viscosity $= \frac{(At - B) \times p}{t}$

Kinematic viscosity $= \frac{u}{P} = \frac{At - B}{t}$

When A and B are constants known as instrument factors, t is Redwood viscosity and p is the density of fluid. The values of A and B for Redwood viscometer No.1 are as follows:-

	A	B
Between 34 – 100 seconds	1.79	0.00260
Above 100 seconds	0.50	0.00247

“Water bath must be filled with water before starting the experiment.

Difference between Redwood No.1 and No. 2:

Redwood viscometer is available in two types, No.1 and No.2. Redwood No.1 and No.2 are similar in construction and working except that the jet of No.1 has a diameter of 1.62mm and length 10 mm. whereas jet of No.2 has a diameter of 3.8mm and length 50mm and Redwood viscometer No.1 is suitable for oils whose time of flow varies from 30 seconds to 2000 seconds and No.2 is suitable beyond 2000 seconds.

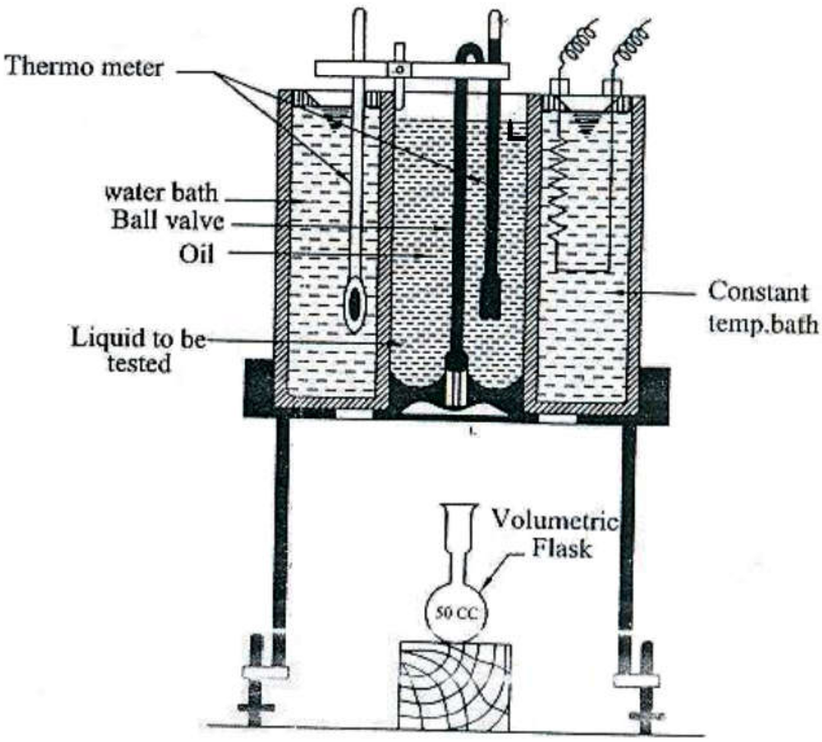

Figure: Redwood Viscometer No. 1 (L. Section view)

1. **The oil cup:** is a silvered copper cylinder which has stout stand. A pointer is fitted inside the oil cup to indicate the level to which the oil is to be filled. The upper end of the cup is open while the bottom of the cup is concave internally and cup has a tapering central hole in which the jet (J) is fitted.

2. **The jet**: constructed of a gate, is provided with a concave depression which can be covered, or uncovered with a ball valve for stopping or starting the flow of oil.

3. **Water bath:** is made up of brass and surrounds the oil cup. It is heated electrically, is provided with a tap for emptying water. Stirring of the bath is affected by means of cylinder surrounding the oil cup and is provided with 4 vanes.

4. **Thermometers** :
   - Two thermometers to read the temperature of oil and water are provided.
   - The apparatus rests on a tripod stand sufficient space for receiving 50 ml. flanged volumetric flask beneath the jet.

Procedure :

1. Viscometer is cleaned and leveled using circular spirit level and base screws the tripod.
2. The ball valve is fixed in position in the oil cup & the oil cup filled up to the level mark with oil.
3. The water bath is than filled with water nearly up to the point of oil level in the cup.
4. The thermometers are inserted in place provided in water bath and oil cup.
5. Now carefully place the 50 ml. measuring cylinder exactly in the centre below the jet.
6. Keep the oil and water well stirred and note their temperature.
7. When the temperature of oil and water becomes same & steady at room temperature, raise the ball valve and suspend it from the thermometer bracket & allow the oil to flow into the measuring cylinder kept below.
8. Simultaneously start a stop watch or look at your wrist watch when the level of oil flowing into the measuring cylinder reaches the 50 ml. mark, note the time in seconds elapsed to collect 50 ml of oil.
9. Replace the ball valve in position to seal the jet to prevent overflow of the oil.
10. Transfer the oil into oil cup from measuring cylinder up to the indicator tip of the cup.
11. Replace properly clean standard measuring cylinder centrally below the jet.
12. The water bath is then heated which in turn heats the oil.
13. When the oil temperature reaches near to the test temperature, stop heating and bring down temperature of water equal to the temperature of oil by running out some hot water through the outlet and adding coldwater from the top with continuous stirring.
14. When temperature of both oil as well as water become steady, again collect 50 ml. of oil and note the time in seconds required.
15. Now repeat the experiment at 5 elevated temperatures. The temperature difference between the two observations should not be less than 5 °C and not more than 10 °C.
16. The temperature of oil and water should always be same while recording the observation.
17. Now plot a graph between temperature and time.

Observation table for Viscometer No.1

S. No	Temperature in °C		Time for collecting 50ml. of oil (in seconds)
	Water	**Oil**	
1.			
2.			
3. 4. 5.			

Calculation:

Absolute kinematic viscosity

$$= \frac{(At - B) \times p}{t}$$

Where A and B are constants known as instrument factors t Redwood viscosity and P is the density of fluid.

Result : Redwood viscosity of the given oil at---**°C** temperature is -----.

The graph between temperature (base) and time (abscissa) in seconds shows that as the temperature increases the time decreases means viscosity of the oil decreases with the increase in temperature.

Precautions:
1. Time should be noted carefully for collecting 50mL of oil.
2. Temperature should be steady and constant while taking observations.
3. Difference in the temperature of water bath & oil should not be more than 2°C while noting time.

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Exercise No. 7

Aim: To determine the viscosity of given sample of lubricating oil with Redwood viscometer No.2 and to study the viscosity at various temperature.

Articles required: Redwood viscometer, thermometers, oil sample, measuring flask 50 ml.

Significance of determination:

Viscosity is the property of a homogeneous liquid which causes it to offer frictional resistance to its motion. Viscosity of lubricating oil tells about its suitability for lubricating purpose.
The function of a lubricant is to reduce the friction between the moving parts and to avoid the direct metal to metal contact. The moving parts remain a part due to extremely thin liquid film in between. The frictional resistance therefore is entirely on account of the shearing of liquid which is in between the metallic surfaces. The viscosity of fluid measures the amount of this internal resistance of lubricating oil.
The object of this test is to ascertain, whether the lubricating oil is sufficiently viscous to adhere to the bearings to which it is applied or is so thin that it may be squeezed out of the bearing due to high pressure & temperature.

Theory/ Principle:

Viscosity is that property of a fluid which is due to the resistance to its flow. It is expressed in 3 ways.
(a) Absolute dynamic viscosity expressed in C.G.S. unit as poise.
(b) Absolute kinematic viscosity expressed in C.G.S. unit as stoke.
(c) Time taken by the determining instrument in seconds noted by the name of instrument (Redwood viscosity).

(a) Absolute dynamic viscosity :

It is the tangential force on unit area of either of two parallel layers at unit distance apart when the space between the planes is filled with the fluid and one of the planes moves relatively to the other with unit velocity in its own plane.
It is defined as the ratio of the shearing stress to the rate of shearing strain. If F is force required to keep moving an article of surface area A in contact with the fluid separated from a stationary surface by thickness d with velocity v then

$$\text{Shearing stress} = \frac{F}{A}$$
$$\text{Rate of shearing strain} = \frac{v}{d}$$
$$\text{Absolute viscosity} = \frac{Fd}{Av}$$

When u is in poise then F = 1 dyne, A = 1 sq.cm. d = 1 cm. v = 1 cm. per second.

(b) **Absolute kinematic viscosity:** It is obtained by dividing Dynamic viscosity by density. The metric units are stoke and centistokes. Thus if p is density, then kinematic viscosity = $\frac{u}{p}$ stoke.

(c) **Redwood Viscosity**: The timing or the rate of flow of given quantity of liquid from a given height through a calibrated capillary tube under its own weight may be used as a means for measuring viscosity and called after the name of the apparatus used for its determination.

The time in seconds required for 50 ml. of oil to gravitate through the jet Redwood viscometer at a given temperature is expressed as its viscosity in Redwood seconds at that temperature.

Absolute dynamic and kinematic viscosities can also be determined from the Redwood values by means of the following formula:

Absolute dynamic viscosity u $= \frac{(At - B)\times p}{t}$

Kinematic viscosity $= \frac{u}{P} = \frac{At - B}{t}$

Where A and B are constants known as instrument factors, t is Redwood viscosity and P is the density of fluid. The values of A and B for Redwood viscometer No.1 are as follows:-

	A	B
Between 34 – 100 seconds	1.79	0.00260
Above 100 seconds	0.50	0.00247

“Water bath must be filled with water before starting the experiment.”

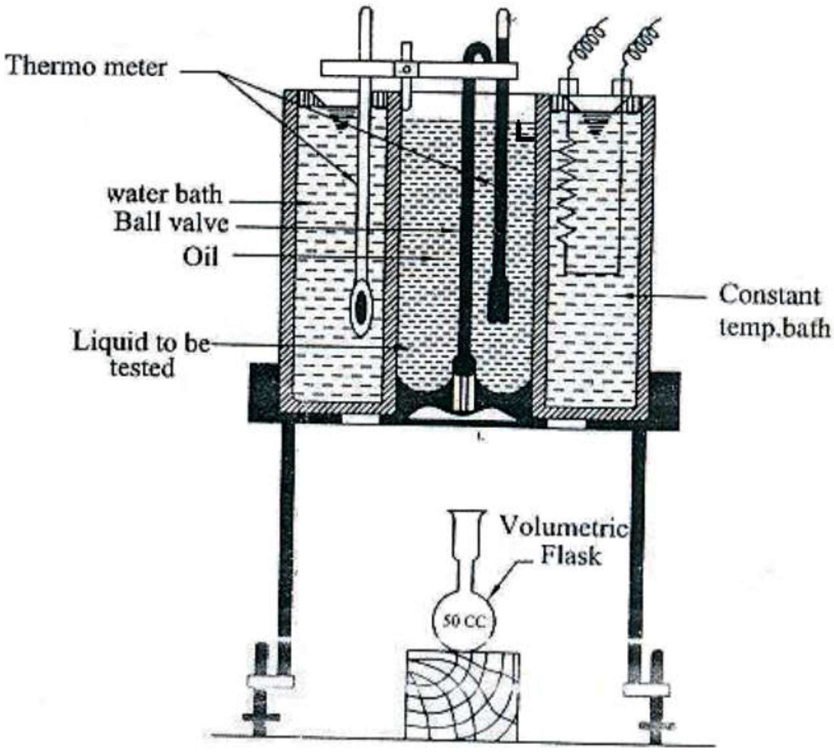

Figure: Redwood Viscometer No. 2 (L. Section view)

Difference between Redwood No.1 and No.2:

Redwood viscometer is available in two types, No. 1 and No.2. Redwood No.1 and No.2 are similar in construction and working except that the jet of No.1 has a diameter of **1.62mm** and length **10 mm**. whereas jet of No.2 has a diameter of **3.8mm** and length **50mm** and Redwood viscometer No.1 is suitable for oils whose time of flow varies from 30 seconds to 2000 seconds and No.2 is suitable beyond 2000 seconds.

The apparatus consists of the following parts.

1. **The <u>oil cup</u>** is a silvered copper cylinder which has stout stand. A pointer is fitted inside the oil cup to indicate the level to which the oil is to be filled. The upper end of the cup is open while the bottom of the cup is concave internally and cup has a tapering central hole in which the jet (J) is fitted.

2. **<u>The jet</u>**, constructed of agate is provided with a concave depression which can be covered, or uncovered with a ball valve for stopping or starting the flow of oil.

3. **<u>Water bath</u>** is made of brass and surrounds the oil cup. It is heated electrically. It is provided with a tap for emptying water. Stirring of the bath is affected by means of cylinder surrounding the oil cup and is provided with 4 vanes.

4. **Thermometers :**
    - Two thermometers to read the temperature of oil and water are provided.
    - The apparatus rests on a tripod stand sufficient space for receiving 50 ml. of measuring cylinder under the jet.
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Procedure:

1. Viscometer is cleaned and leveled using circular spirit level and base screws the tripod
2. The ball valve is fixed in position in the oil cup & the oil cup filled up to the level mark with oil.
3. The water bath is than filled with water nearly up to the point of oil level in the cup. The thermometers are inserted in place provided in water bath and oil cup.
4. Now properly place the 50 ml. measuring cylinder exactly in the centre below the jet.
5. Keep the oil and water well stirred and note their temperature before heating.
6. When the temperature of oil and water becomes same & steady raise the ball valve and suspend it from the thermometer bracket &allow the oil to flow into the measuring cylinder kept below.
7. Simultaneously start a stop watch or look at your wrist watch when the level of oil flowing into the measuring cylinder reaches the 50 ml. mark, note the time in seconds elapsed to collect the 50 ml oil at room temperature.
8. Replace the ball valve in position to seal the jet to prevent overflow of the oil.
9. Transfer the oil into oil cup from measuring cylinder up to the indicator tip of the cup.
10. Properly replace clean standard measuring cylinder centrally below jet.
11. The water bath is then heated which in turn heats the oil.

12. When the temperature of oil reaches near to the test temperature stop heating and brings down the temperature of water equal to it by running out some hot water through the outlet and adding coldwater from the top while stirring continuously.
13. When temperature of both oil as well as water become steady and constant, again collect 50 ml. of oil and note the time in seconds required to collect 50 mL of oil.
14. Now repeat the experiment at 5 elevated temperatures. The temperature difference between the two observations should not be less than 5 °C and more than 10 °C.
15. The temperature of oil and water should always be same, steady and constant while recording the each observation.
16. Now plot a graph in between temperature and time.

Observation table: for Viscometer No.2

S. No	**Temperature in °C**		**Time in seconds for flow of 50ml. of oil**
	Water	**Oil**	
1.			
2.			
3. 4. 5.			

Calculation:

Absolute kinematic viscosity

$$= \frac{(At - B) \times p}{t}$$

Where A and B are constants known as instrument factors of Redwood viscosity and P is the density of fluid.

Result: Redwood viscosity of the given oil at---**°C** is -------

The graph between temperature (base) and time (abscissa) in seconds shows that as the temperature increases the time decreases means viscosity of the oil decreases with the increase in temperature.

Precautions:

1. Time should be noted carefully every time during collection of 50ml. of oil.
2. Temperature should be steady while noting time for flow.
3. Difference in the temperature of water bath & oil should not very even by 1°C while noting time.

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Exercise No. 2.

Aim : To determine flash point of illuminating or lubricating oil by Abel's apparatus.

Articles required : Abel's apparatus, thermometers, oil sample, water and LPG etc..

Definition : Flash point of oil is defined as the minimum temperature at which it gives off sufficient vapours, which when mixed with air forms an ignitable mixture and gives a momentary flash of light on the application of a small flame.

Significance of determination : The determination of flash point is of great importance in case of lubricating and illuminating oils. If the oils are sufficiently volatile at, ordinary temperatures, the issuing vapors will form an explosive mixture with air and cause fire hazards. To ensure safety while using oils, certain minimum temperatures are laid down for illuminating and lubricating purposes, below which it should not give off inflammable vapors.
The flash point of oil shows the extent of more volatile constituents in the oil. The lower the flash point the greater is the amount of more volatile constituents.
The flash point is clearly distinguished from fire point, which is the temperature at which the air space above the oil has sufficient concentration of vapors so that it burns continuously for about five seconds when brought in contact with the flames.

Part of Apparatus : The Abel's apparatus is a cylindrical structure consisting of the following parts :
1. **The oil cup :** This is a cylindrical vessel, made of brass, open at the top end fitted on the outside with a flat circular flange. Projecting at right angles, within the cup there is a gauge consisting of a piece of wire bent upwards and terminating in a point. The oil is filled up to this point.
2. **Cover :** This is fitted with a stirrer and thermometer socket made of brass mounted on the oil cup.
3. **Heating Vessel :** The heating vessel consists of two flat bottomed cylindrical vessels, placed coaxially one inside the other and soldered at their tops to a flat copper ring. The space between the two vessels thus totally enclosed and is used as a water jacket.
In addition to above the working with Abel's apparatus requires the heating arrangement and thermometers of appropriate range. It may be heated electrically or with a burner.

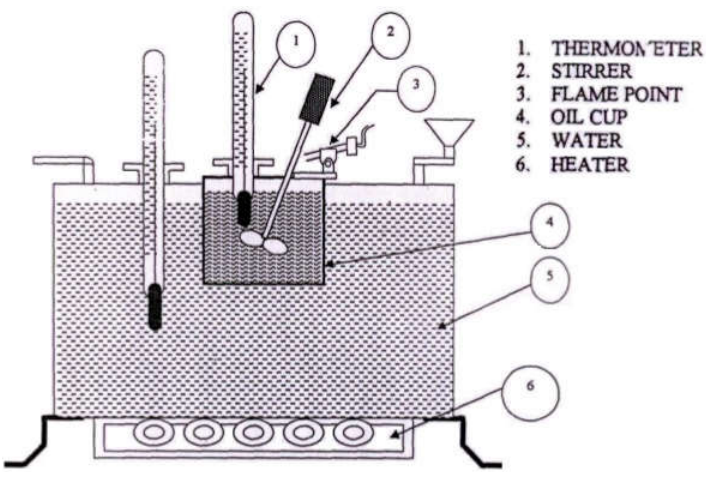

Figure: Abel's Apparatus (L. Section view)

Procedure :

1. Fill the oil cup with the oil under test up to the indicator mark. Replace the cover.
2. Fix the oil cup into the apparatus and assemble the paddle stirrer and thermometers with its bulb dipping into the oil and water at their respective places, provided for in the apparatus.
3. Fill the water bath with cold water.
4. Close the sliding shutter and light the standard flame and start heating the water bath with the help of burner.
5. Stir the oil continuously by turning the paddle stirrer. Stirring should be discontinued only during the introduction of the test flame over the oil surface.
6. At every degree rise of oil temperature open the sliding shutter and introduce the test flame over the oil surface through the central opening to see whether the oil gives a flash or not.
7. Record the minimum temperature at which a distinct flash appears as the flash point of the given oil.
8. Now cool the oil by 5°C and verify the flash point again by every 1°C rise of temperature.

Observation table No. 1:

S. No.	Temperature (°C)	Observation
1.	Room temperature	No flash
2.		
3.		
4.		
5.		
6.		
7.		
49.	49	Flash

		observed

Observation table No. 2: After cooling of 5 °C.

S. No.	Temperature (°C)	Observation
1.		No flash
2.		
3.	47	Flash observed
1.		No flash
3.	47	Flash observed

Result: The flash point of given oil sample is --------- °C

Precautions :

1.
2.
3.
4.
5.

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Exercise No. 6

Aim : To determine the flash point of lubricating oil by Pensky Martens apparatus.

Articles required : Pensky Martens apparatus, thermometers, oil sample and LPG etc.

Definition : Flash point of oil is defined as the minimum temperature at which it gives off sufficient vapors, which when mixed with air forms an ignitable mixture and gives a momentary flash of light on the application of a small flame.

Significance of determination : The determination of flash point is of great importance in case of lubricating and illuminating oils. If the oils are sufficiently volatile at ordinary temperatures, the issuing vapors will form an explosive mixture with air and cause fire hazards. To ensure safety while using oils, certain minimum temperatures are laid down for illuminating and lubricating purposes, below which it should not give off inflammable vapors.

The flash point of oil shows the extent of more volatile constituents in the oil. The lower the flash point the greater is the amount of more volatile constituents.

The flash point is clearly distinguished from fire point, which is the temperature at which the air space above the oil has sufficient concentration of vapours so that it burns continuously for about five seconds when brought in contact with the flames.

Part of Apparatus : The Pensky Martens apparatus consists of the following main parts.

1. **The Oil Cup :** It is a cylindrical vessel, made of brass with a filling mark grooved inside near the top. The cup is covered with a lid.

2. **Lid : T**he lid is equipped with the following parts:

(i) **Stirrer :** The stirring device is carrying two bladed brass propellers mounted in the centre of the cup.

(ii) **Cover Proper :** It is made of brass and is provided with four openings one for thermometer and the rest for the oxygen entry and exposure of vapours to test flame.

(iii) **Shutter :** It is made of brass and is so shaped and mounted on the lid that when in one position the holes are completely closed and when in the other these openings are completely opened so as to test the material for its flash point. It can be worked by rotating the knob.

(iv) **Flame Exposure Device :** The lid is provided with a pilot lamp with such a mechanism that its flame operates simultaneously with the shutter. When the shutter is in the ‘open’ position the tip of the flame is lowered down

(v) **Heating Device :** The oil cup is placed in another cup of slightly bigger dimensions. Air is present between the two cups thus providing an air bath which is heated electrically.

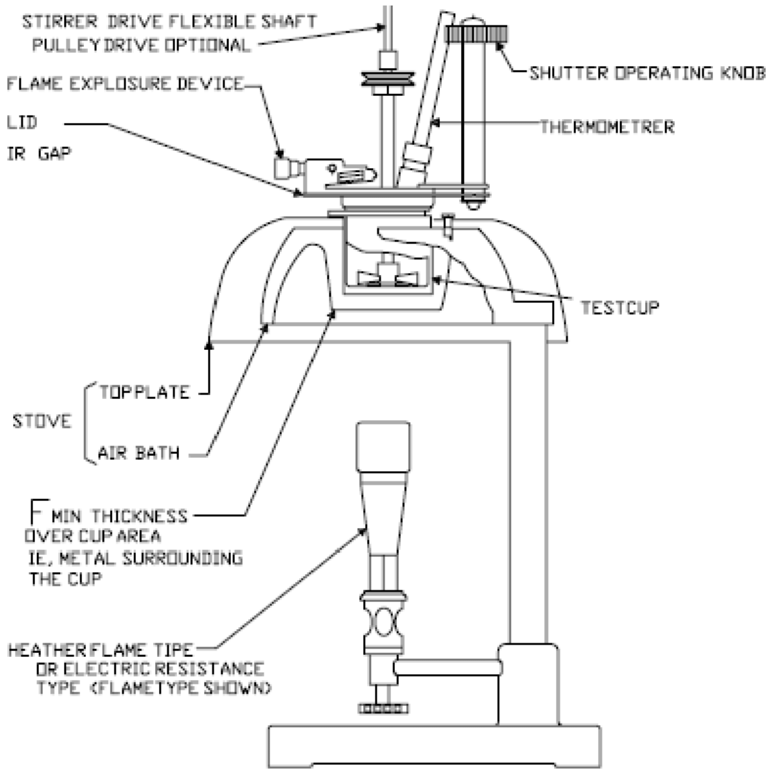

Figure: Pensky Martens Apparatus (L. Section view)

Procedure :

1. The oil cup and its accessories are cleaned and dried before the test is performed.
2. Fill the oil cup with already dehydrated oil sample to be tested up to the filling mark (A groove cut inside the oil cup). Then place the cup in the heating bath.
3. Replace the cover of the oil cup.
4. Insert the thermometer into the thermometer socket, provided on the cover proper.
5. Heat the oil cup from below with burner with constant slow stirring.
6. After every 10°C rise of temperature upto 100 °C and therafter every 5°C rise of temperature, open the shutter by operating knob and introduce test flame over the oil surface through the central opening to see whether the oil gives flash or not.
7. When the test flame causes a distinct flash in the interior of the cup, note temperature which represents the flash point.
8. Place the oil cup in the tray and allow it to cool down by10 to 15°C start heating the sample again and test for the flash point after every 1°C rise in temperature till flash observed.
9. Again cooldown the oil cub by 2 to 5 °C then check flash point by every 1°C rise of temperature till flash observed.
10. Repeat the last step until two consecutive concorrent temperature.

Observation Table No.1 : Every 10 °C rise of temperature.

S.No.	**Temperature (°C)**	**Observation**
1.	Room temperature	No flash
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.	120	Flash observed

Observation Table No.2 : Every 1°C rise of temperature.

S.No.	**Temperature (°C)**	**Observation**
1.	**115**	No flash
2.		
3.		
4.	**118**	Flash observed

Observation Table No.3 : Every 1°C rise of temperature.

S.No.	**Temperature (°C)**	**Observation**
1.	**117**	No flash
2.	**118**	Flash observed
3.	**117**	No flash
4.	**118**	Flash observed

Result: The flash Point of given oil sample is ……….°C

Precautions :

1.

2.

3.

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Exercise No. 3

Aim : Determination of moisture content of coal or coke sample.

Articles required: Clean and dry china dish, chemical balance, hot air oven, pair of tongs, desiccator weight boxes (major & fractional upto the weight of 10mg), ridder, forcep,etc.

Coal : It is most import natural solid fuel used for domestic as well as industrial purposes. It is highly carbonaceous matter containing carbon, hydrogen, nitrogen and oxygen besides non-combustible inorganic matter.
Quality of coal is ascertained by two types of analyses:

1. **Proximate analysis** involving determination of :
   (a) Moisture
   (b) Volatile matter
   (c) Ash content and
   (d) Fixed carbon

2. **Ultimate analysis** involving determination of :
   (a) Carbon and Hydrogen
   (b) Nitrogen
   (c) Sulphur and
   (d) Ash content

Significance of determination :

(A) Coal samples contain natural moisture and it may increase during
(i) cleaning of coal by gravity or flotation separation
(ii) storage of coal under water to avoid spontaneous ignition causing fire hazards.
Coal are recognised or selected for a particular purpose on the basis of rank or degree of qualification and transformation from wood to Peat to Lignite toBituminous and finally into Anthracite. This results in :
1. Increase in carbon content, hardness, calorific value.
2. Decrease in oxygen content, moisture and volatile matter.
Coal with low moisture belongs to a higher rank, possesses higher calorific value and sold / purchased as a superior grade fuel.

(B) Moisture involves a loss of money paid for it at the rate of coal during purchase and transportation.

Description :
Chemical balance : It is normally used to weigh accurately to the fourth place of decimal using a rider.
Air oven : Electrically heated air oven in which temperature can be maintained to the accuracy of +2°C.
Desiccator : Itis a glass container used to dry chemicals or keep them dry. Generally anhydrous calcium chloride or conc. Sulphuric acid is used as moisture absorbent.

Procedure :
Take a clean dried crucible and weigh it accurately to the fourth place of decimal (using a rider). Now take approximet 1 gm of coal sample in powdered form and weigh again. Using tongs put the weighed crucible containing coal sample in a desiccator and take the desiccator to the oven. Transfers the crucible carefully with the help of pair of tongs to the oven maintained at 115^oC and heat the sample for 10 – 15 minutes in the oven. Take out the crucible containing sample from the oven with tongs and keep it in the desiccator (closed) for cooling to room temperature for about 2-5 minutes and weigh it to find out the loss of moisture.

Observation :

1. Weight of the empty crucible = w gm.
2. Weight of the crucible + coal sample = w1 gm.
3. Weight of coal sample taken = $(w_1 - w)$ gm.
4. Weight of the crucible + coal = w2 gm. After heating at 115^oC.
5. Weight of coal sample after moisture removal = $(w_2 - w)$
6. Loss in weight = $(w_1 - w) - (w_2 - w)$
   = $(w_1 - w_2)$ gm.

Calculation and Result : Percentage of Moisture in the coal sample

$$= 100 \times \frac{(w_1 - w_2)}{(w_1 - w)}$$

Conclusion : The moisture content in the sample being

(a) Too high – sample needs drying before use.
(b) Normal – sample can be used as such.
(c) Quite low – sample is very good.

Precautions :

(i) The coal sample should be weighed exactly to the fourth place of decimal.
(ii)The windows of the chemical balance should be kept closed during weighing operation.
(iii) The crucible should be handled with a pair of tongs.
(iv) The coal sample should be placed in a desiccator for cooling.

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Exercise No. 4

Aim : To determine Steam Emulsification Numbers of an oil sample.

Articles required : Thermometers, oil sample, SEN apparatus & LPG.

Definition : The time in seconds elapsed during the separation of 20 mL. of oil from approximately double of its emulsion with water is reported as steam emulsification number.

Significance of determination : Certain lubricating oils mix intimately with water and form emulsion, which then has a tendency to collect dirt, grit particles and other foreign matters. Thus solid particles are introduced into the lubricated parts and cause abrasion and wearing out of the machinery.

The tendency of oil to emulsify increases with the time it is used as lubricant. Good lubricating oil is one which does not form emulsion under lubricating conditions and even if it forms, it should break off very quickly.

The tendency of emulsion is determined by A.S.T.M. emulsion test. An emulsion of oil and water is made under standard conditions and then allowed to break. **The time in seconds required for the complete separation of water from emulsion is reported as SEN.**

In circulating oiling systems, the high SEN is particularly undesirable because sticky emulsions and sludge interfere with the circulation of oil. But in certain steam cylinder oils, the formation of stable emulsion is desirable. In steam turbines, the oils should separate quickly so that it can be re-circulated for use.

SEN or RE (Rate of emulsion separation) is directly proportional to the emulsification of tested oil.

To establish a basis for comparison a definite value 100 is assigned to an oil which under condition of test separates completely from emulsion in one minute. The rate is 33 cc per second or 20 cc per minute. It was found after experimentation that the results were pertinent and relative through a wide range of tests. The value assigned is denoted RE or resistance to emulsion.

The RE of any oil may be found by substituting in following formula

$$\text{RE} = \frac{cc\ of\ separated\ oil \times 5}{Number\ of\ minutes}$$

Parts of apparatus:

(i) **Steam generator 'C' :** It is a round bottomed flask heated from below with the burner.

(ii) **Test tube:** A 2.5 by 20 cm test tube 'D' used as a sample tube graduated (in some) from zero to 50 ML

(iii) **An emulsifying bath 'B':** It is a glass battery jar of 3 litre capacity.

(iv) **Separating Bath 'F':** It is a glass battery jar of 3 litre capacity.

(v) **Delivery Tube 'D':** Steam from 'C' is led into the sample by means of a delivery tube the other end of which is cut at an angle of 30^{o}C.

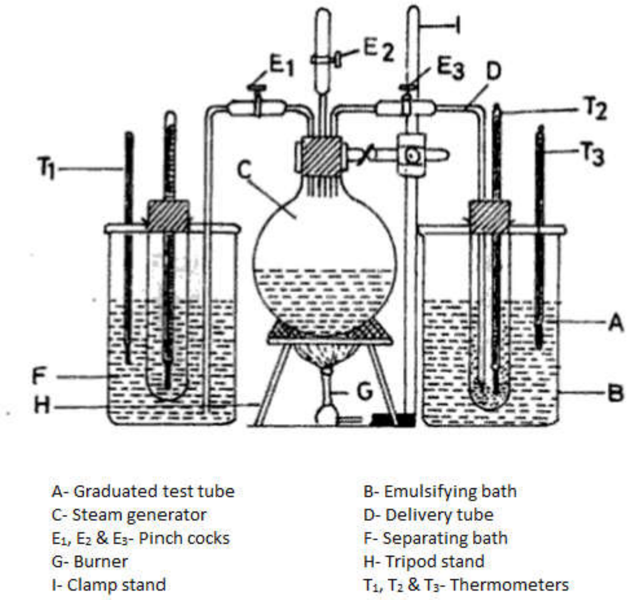


Figure:Steam Emulsification Apperatus

Procedure:

1. Take approximately 20 ml. of the oil sample in a test tube. Adjust the thread mark on the test tube up to 40 ml.
2. Place the test tube into the beaker kept in tray (half filled with water).
3. Now check that steam generator is filled with water (nearly half of its capacity).
4. Fill the other beaker 3/4 with water.
5. Now heat this beaker as well as steam generator from below with the help of burners separately.
6. The temperature of water in beaker should be less than 100^{o}C (as soon as some vapours form on the water, stop heating).
7. When a continuous steam is coming out from the delivery tube introduce it is the test tube containing oil sample (nearly up to the bottom).
8. Continue passing steam until the content of the test tube becoms double (reachs up to the thread mark).
9. Immediately remove the steam delivery tube and transfer the test tube into the beaker (containing hot water) along with the cover.
10. Note down the temperature in seconds required for complete separation of water and oil as SEN.
11. Now transfer the contents of the test tube into the emulsion bottle kept in the tray.
12. Repeat this exercise thrice.

Observation Table:

S.No.	**SEN**
1.	
2.	
3.	
4.	
5.	

Result: Steam emulsification No. of the given oil is …………..

Precautions:

1. Steaming rate should be controlled as described.
2. After emulsification the test tube should be transferred rapidly to the separating bath.

****

Exercise No. 10

Aim : To determine the acid no. of given oil.

Articles required :
Apparatus : Conical flask, Measuring cylinder, Burette, Water Bath,Air Condenser,etc.
Chemicals : Deionized water, Ethanol, oil sample, Phenolphthalein indicator,etc.

Significance of Determination : The value is a measure of the amount of fatty acids, which have been liberated by hydrolysis from the glycerides due to the action of moisture, temperature and/or lipase (lipolytic enzyme) .

Theory/ Principle : The acid value is defined as the number of milligrams of Potassium hydroxide required to neutralize the free fatty acids present in one gram of fat. It is a relative measure of rancidity as free fatty acids are normally formed during decomposition of triglycerides. The value is also expressed as per cent of free fatty acids calculated as oleic acid, lauric, ricinoleic and palmitic acids. The acid value is determined by directly titrating the oil/fat in *an alcoholic medium* against standard alkali solution like potassium hydroxide/sodium hydroxide solution.

Procedure :
1. Weigh accuretly 5 gms of the cooled oil sample in a 250 mL conical flask.
2. Add 10 mL of distilled water.
3. Add 10 mL of freshly neutralised ethyl alcohol.
4. Add about 1-2 drops of phenolphthalein indicator solution.
5. Heat the mixture for about five minutes in water bath (75-80°C).
6. Titrate carefully while hot against standard alkali solution shaking vigorously till solution becoms light pink from colorle.

Calculations :

$$\text{Acid value} = \frac{56.5 \times V \times N}{W}$$

Where,
V = Volume in mL of standard potassium hydroxide or sodium hydroxide used.
N = Normality of the alkali solution (potassium hydroxide or Sodium hydroxide solution).
W = Weight in gm of the sample.

Result: Acid no. of given oil is --------------(mg/KOH/g oil)

Precautions:
1. Clamp conical flask proprerly.
2. Keep alcohol and water mixture away from burner.
3. Use water bath for heating.
4. Condenser should fitted tightly.

****

Exercise No. 5

Aim : To determine the total solids in water (Dissolved and suspended).

Articles required : Clean and dry china dish, Chemical balance, Hot air oven, Pair of tongs, Desiccator,etc.

Object and principle : High concentration of dissolved solids in boiler leads to foaming and priming inside the boiler. Various other difficulties are being encountered in normal and regular operation of boiler. Removal of solids deposited will necessitate a great number of boiler blow downs.

The quantitative determination of solids in boiler feed water is very essential. When a known volume of water sample is evaporated to dryness it gives rise to residue. This residue constitutes the solids present in the water sample.

Procedure :

Weigh out an evaporating china dish . Take 100 mL of water sample in it. Evaporate water sample to dryness carefully. When it is almost dried, keep the china dish in an oven at 100-120^{o}C for 2-5 minutes. See that the residual water is completely removed. Cool the china dish in a desiccator to room temperature. Find the weight of residue.

Observations :

Volume of water sample = 100 ml.
Weight of empty china dish = x gms.
Weight of residue + china dish = y gms.
Weight of residue = y – x gms.

Calculations : Total solids present in water sample.

$$= (y - x) \text{ gm/100 ml.}$$
$$= (y - x) \times 10 \text{ gm/litre}$$
$$= (y - x) \times 10^{-4} \text{ gm/litre}$$
$$= (y - x) \text{ gm/} \times 10^{-4} \text{ ppm.}$$

Result : The amount of total solids in the given sample of water is found to be ………… ppm.

****

Exercise No. 6

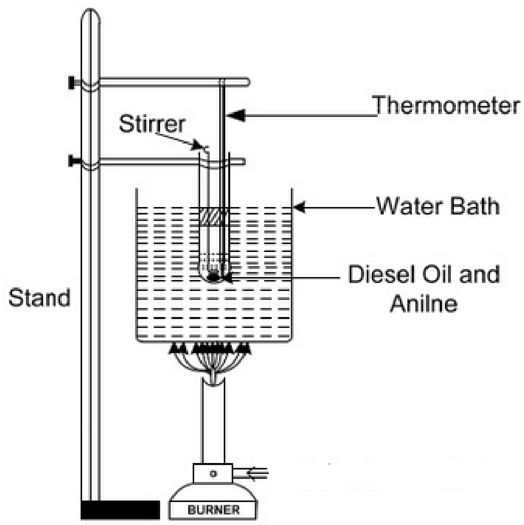
Aim : To determine the Aniline Point of the given oil sample.

Articles required : Graduated test tube, Aniline point apparatus, Thermometer, water bath, and stirrer, etc.

Definition : Aniline point may be defined as the lowest temperature at which the oil is completely miscible with an equal volume of aniline.

Significance of Determination : Determination of aniline point of fuel oils is very informative. If petroleum or mineral fuel oils contain a higher percentage of straight chain hydrocarbons, it produces a rattling noise in the internal combustion engine called knocking. This knocking is minimum, if the percentage of branched chain hydrocarbons and or aromatic hydrocarbons is higher. Determination of aniline point gives information about the type of hydrocarbon content of the fuel oil.
If, the aniline point is lower, the proportion of aromatic hydrocarbons is higher and the oil will have less or no tendency of knocking in the engine.Aromatic hydrocarbons present in lubricants have the tendency to dissolve rubber or synthetic rubber packings, sailings etc; therefore, aniline point of oil gives indication regarding the deterioration, dissolution of rubber sealing, packing etc. in contact with lubricating oils. For good lubricating oil the aromatic hydrocarbon content should be low. The higher the aniline point, the lower is the percentage of aromatic hydrocarbons in the fuel or lubricating oil and is safer to use in contact with rubber parts. Aniline point gives valuable information about the presence of type hydrocarbon in the oil.

Parts of apparatus : The aniline point apparatus consists of;
1.Test tube : It is to hold the sample. It is made of heat resistant glass and is approximately 25mm in diameter and 150mm in length. It is fitted with a cork bored to hold the thermometer centrally and provided with another hole through which stirrer can operate freely. This tube is suspended in a jacket, made of similar glass material. The test tube along with jacket is kept immersed in a heating or cooling bath.
2.The stirrer : It is made of metal or glass rod bent at the bottom into a ring about 15mm in diameter at right angle to the main axis.
3. **The thermometer :** is suspended in the test tube through a central hole in the cork.



LABORATORY ANILINE POINT APPARATUS

Procedure :

1.Equal volume of pure and anhydrous aniline and oil are taken in the test tube (two layers are observed).
2.Water bath is heated and test tube containing oil and aniline is placed in it.
3.The contents of test tube are stirred (by up and down motion of stirrer) till a homogenous solution is obtained.
4. Thermometer is placed in the test tube and temperature is recorded.
5. The test tube is then placed in the cooling jacket of the aniline point apparatus.
6. Initially cloudiness appears in the solution and then two layers are obtained.
7.The temperature at which cloudiness appears is noted.
4. The lowest temperature at which cloudiness appears is reported as aniline point.
5. Procedure is repeated thrice, till two concorrent results are obtained.

Observation :

Volume of oil taken = V ml.
Volume of aniline taken = V ml.
(Equal volume)

<u>Observation Table :</u>

S.No.	Temperature at which cloudiness appears in homogeneous solution (°C)
1	
2	
3	

Lowest value = ………..°C

Result and conclusion : The aniline point of the given lubricating oil is ………°C

1. If the aniline point is low, the percentage of aromatic hydrocarbons is higher and the oil should not be used in contact with rubber.
2. If the aniline point is higher, the percentage of aromatic hydrocarbons is lower and oil is suitable for use in contact with rubber.

Precautions :

1. Pure aniline should be used.
3. Stirring should be done by up and down motion of stirrer with care. No part of the mixture should be allowed to splash out.
4. Aniline point should be recorded with care and experiment should be repeated at least 2 times for consistency

****

Exercise No. 10

Aim: Determination of drop point of a grease sample.

Articles required: Drop point apparatus, water bath, semi solid lubricant(grease), and thermometer,etc.

Significance of Determination: Drop point is the temperature at which the grease passes from semi-solid state to the liquid state. It determines the upper limit of temperature up to which grease can function as a satisfactory lubricant.

Parts of apparatus:
It consists of a metal cup having an opening of a standard dimension at the bottom. The cup is fitted in a glass jar(boiling tube) with a tightly fitting lid. A thermometer is fixed in the cup that the bulb of the thermometer is just above the surface of the grease sample under test. The entire assembly is held in position in a glass beaker containing water which is also provided with a thermometer & a stirrer.

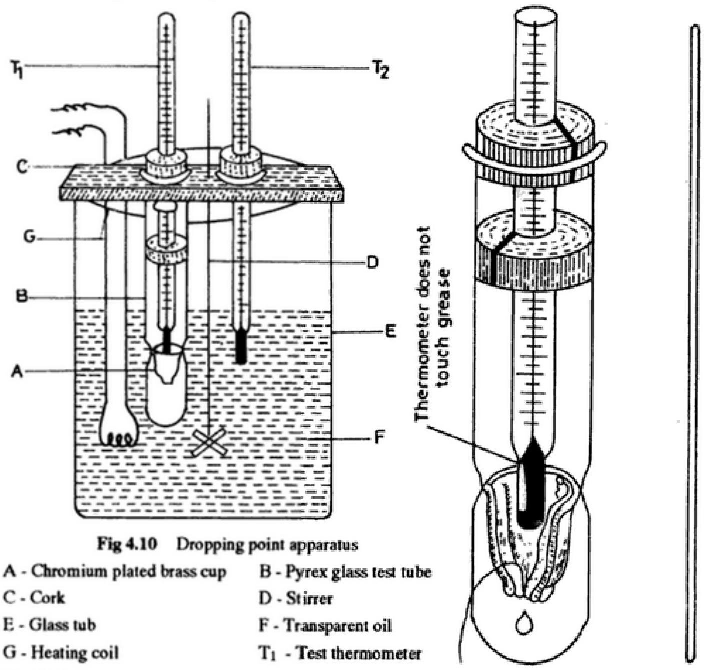


Fig 4.10 Dropping point apparatus

A - Chromium plated brass cup
B - Pyrex glass test tube
C - Cork
D - Stirrer
E - Glass tub
F - Transparent oil
G - Heating coil
T_1 - Test thermometer
T_2 - Bath thermometer

Drop Point Apparatus

Procedure:
1. Place the grease sample in the cup and insert the thermometer in place.
2. Fix the cup along with thermometer into a boiling tube containing cotton. Now place the boiling tube into the beaker containing water.
3. Heat the beaker from the below with burner slowly. As the temperature increases the grease sample melts and the first drop emerges out of the opening at the bottom of the metal cup.

4. Record the temperature at which the first drop falls down. This temperature is noted as drop point.
5. Now remove the burner.
6. Take out small amount of hot water (20 – 25 ml.) and mix cold water.
7. The temperature of water should fall down by 10 °C.
6. Replace the test tube into the beaker and repeat the experiment 2-3 times, till two similar readings are

Observation table

S.No.	Drop Point (°C)
1	
2	
3	

Result : The drop point of given grease sample is …………. °C

Precautions :
1. Pure aniline should be used.
2. Drop point should be recorded with care and experiment repeated at least 2 times for consistency.

****

Exercise No. 11
SYNTHESIS OF POLYMER AND NANOMATERIALS

PART I

Aim : To prepare Phenol formaldehyde (P-F) resin.

Articles required :
Apparatus : Glass rod, beakers, funnel, measuring cylinder, dropper and filter paper.
Chemicals: Phenol, aq. formaldehyde solution or formalin, glacial acetic acid and conc.HCl, etc.

Significance of preparation: Bakelite is condensation type of thermosetting polymer. This polymer is applied in many task of our routine life, few uses of this polymer areas follows- used for making molded articles such as radio and TV parts, combs, fountain pen , barrels, phonograph records, decorative laminates, wall coverings, electrical goods such as switches, plugs etc, for impregnating fabrics ,wood and paper, for bonding glue, for laminated wooden planks, in varnishes and lacquers. Sulphonated phenol-formaldehyde resins are use as ion-exchange resins.

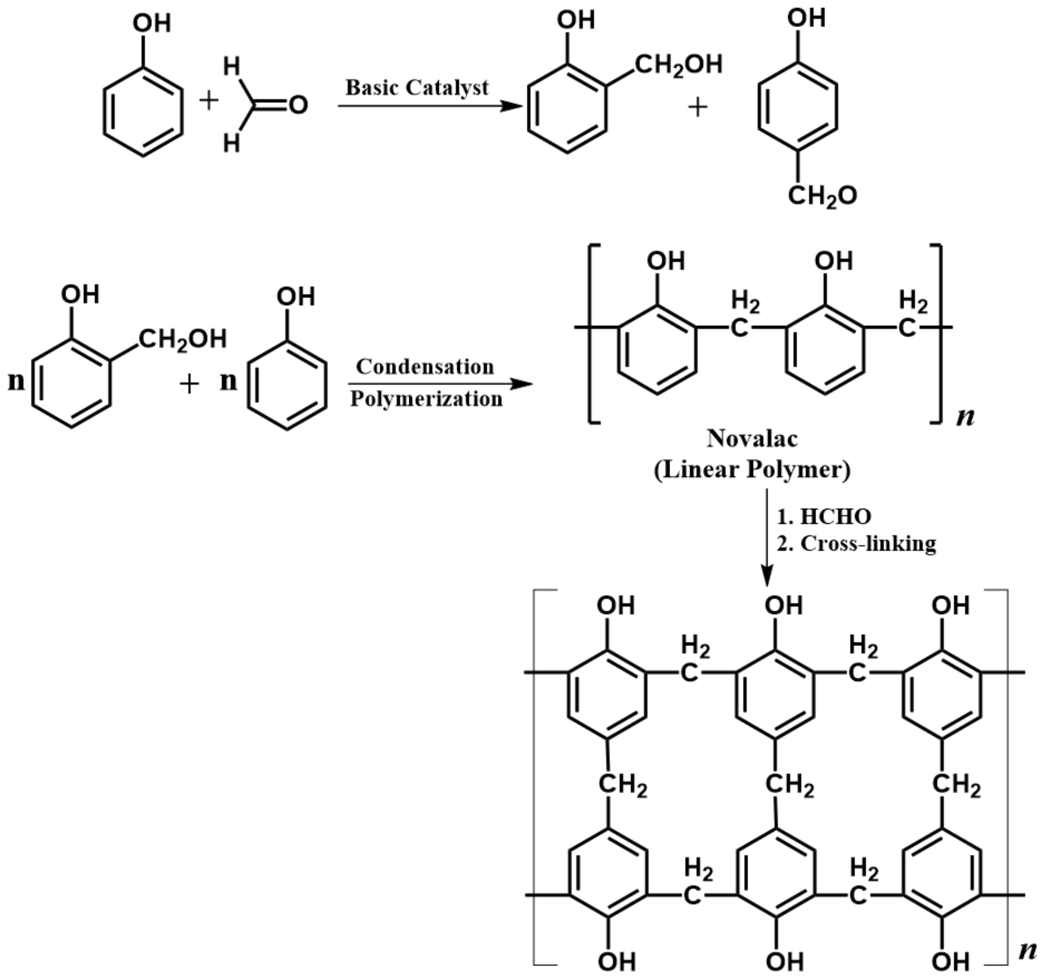
Theory/ Principle: Phenol formaldehyde resins (PFs) are condensation polymers and are obtained by condensing phenol with formaldehyde in the presence of an acidic or alkaline catalyst. They were first prepared by Baekeland, an American Chemist who gave them the name as Bakelite.

Thermo sets: The polymers which on heating change irreversibly into hard rigid and infusible materials are called thermosetting polymers. These polymers are usually prepared by heating relatively low molecular mass, semi fluid polymers, which becomes infusible and form an insoluble hard mass on heating. The hardening on heating is due to the formation of extensive cross-linking between different polymeric chains. This leads to the formation of a 3-dimnesional network of bonds connecting the polymer chains. Since the 3D network structure is rigid and does not soften on heating, the thermosetting polymers cannot be reprocessed. Some important examples of thermosetting polymers are Urea-Formaldehyde resin and Melamine-Formaldehyde resins.

Properties:
1. Phenol- formaldehyde resins having low degree of polymerization are soft. They possess excellent adhesive properties and are usually used as bonding glue for laminated wooden planks and in varnishes and lockers.
2. Phenol- formaldehyde resins having high degree of polymerization are hard, rigid, scratch- resistant and infusible. They are resistant to non-oxidizing acids, salts and many organic solvents. They can withstand very high temperatures. They act as excellent electrical insulators also.

Procedure:
1. Place 5mL of glacial acetic acid and 2.5mL of 40 % aq. Formaldehyde (formaline) solution in a100 mL beaker. Add 2g phenol safely.
2. Wrap the beaker with a wet cloth or place it in a 250mL beaker having small amount of water in it.
3. Add conc. HCl drop wise with vigorous stirring by a glass rod till a pink colored gummy mass appears.

4. Wash the pink residue several times with to make it free from acid.
5. Filter the product and weigh it after drying in folds of a filter or in an oven. Report the yield of polymer formed.



Scheme: Bakelite synthesis

Observations:

Weight of empty watch glass = W1 g
Weight of watch glass + polymer formed = W2 g
Weight of polymer formed = W2 – W1 g

Result:

Weight of phenol formaldehyde resin = W g

Precautions:

1. The reaction is sometimes vigorous and it is better to be a few feet away from the beaker while adding the H_2SO_4 and until the reaction is complete.
2. The experiment should be preferably carried out in fume chamber.

****

PART II
Aim: To synthesize silver nanoparticls.

Articles required:
Apparatus: Beaker, measuring cylinder, magnetic stirrer, magnetic bar, etc.
Chemicals: Silver nitrate, sodium borohydride, trisodium citrate and distilled water.

Significance of preparation: Silver nanoparticles are important in the field of medicine as they show strong antibacterial properties.

Theory/ Principle: Silver can be reduced from Ag^+ to Ag^0 by various reducing agents. One of the common reducing agents is sodium borohydride ($NaBH_4$) as it is cheap and easily available. $NaBH_4$ reduces the silver ions to silver atoms which cluster together to form nanoparticles. The citrate salt acts as capping agent and stabilizes the silver nanoparticles, thus preventing their aggregation to larger particles.

Procedure:
1. 15mL of 0.002 M sodium borohydride is taken in a beaker in an ice cold water bath. To this, 5mL of 0.001 M sodium citrate is added and the mixture is stirred using a magnetic stirrer.
2. Now add 2mL of 0.001 M $AgNO_3$ solution to the above solution in a drop wise manner.
3. Stir the solutions for 15 minutes.

Result: The formation of a yellow colored solution indicates the presence of formation of silver nanoparticles.

Precautions:
Perform the reaction in an ice cold water bath as the reduction of silver ions by sodium borohydride is a fast reaction and can lead to generation of larger particles at higher temperature.

****

Exercise No. 12

Aim : To verify Beer - Lambert's law by UV -Visible spectroscopy.

Articles required :
Apparatus : Beaker, Measuring cylinders, Volumetric flask and Cuvette,etc.
Chemicals : Distilled water and $KMnO_4$,

Significance of determination: To study the dependence of absorbance on the concentration and path length and confirm the applicability range. Every absorbing species has a characteristic λ_{max} value which is determined by this method.

Theory:
Absorption spectroscopy can be used to quantify the absorbing species present in a sample. The greater the quantity of absorbing species present, greater will be the extent of absorption of incident light through the solution or gasses. When a beam of monochromatic light falls on a substance, a part of it is absorbed and the rest is transmitted. The intensity of the transmitted light is decreased.
According to Lambert's law, decrease in intensity (-dI) is proportional to the thickness of the medium (dl) and intensity of incident light (I). Beer extended this law for solutions and gases. In such cases decrease in intensity is also proportional to molar concentration (C).

$$-dI \propto I.dI.c$$
$$-dI = k.I.dI.c$$

Where, k = constant
For a solution, the absorbance of the sample (A) is given by:

$$A = log\frac{I_0}{I} = \varepsilon.c.l$$

Where, I_0 = intensity of the incident light,
I = intensity of the transmitted light,
ε = molar absorptivity (characteristic of the absorbing species),
c = molar concentration of the absorbing species in the sample,
l = distance that the incident radiation travels through the sample.
$log\frac{I_0}{I}$ = Optical density or Absorbance

Since 'ε′ depends on wavelength of the light and nature of the substance so for a given substance at a given frequency and in a given photochemical cell.
Beer - Lambert's law states that when l = constant, O.D. = constant $*c$. then the plot of O.D. Vs C will be a straight line through origin.
Absorbance has no units. If $'l'$ is expressed in cm and ' c ' in mol/L, then the unit of ' ε ' will be $Lmol^{-1}cm.^{-1.}$
A linear plot of absorbance versus concentration would verify the Beer-Lambert's Law.

Procedure:
1. 100 ml stock solution of 0.001 M $KMnO_4$ is to be prepared.
2. Know standard solution of 0.0001 M $KMnO_4$ can be prepared by pipeting out 10 ml of stock solution into 100 ml standard flask and make up upto the mark with distilled water.
3.Record the visible spectrum of 0.0001 M $KMnO_4$ solution in the wavelength range of 350 nm to 800 nm, at the intervals of every 10 nm to get maximum wavelength i.e., λ_{max}. Note your results in table 1.

4. A plot of Absorbance 'A' vs. wavelength in nm gives λ_{max}.
5. From the standard 0.0001 M$KMnO_4$ solution, Prepare 10ml solutions of each proportion of $KMnO_4$ solution as given in the table 2.
6. Measure the absorbance 'A' at λ_{max} for each of the solutions prepared in Step 5 and then for the unknown solution of $KMnO_4$ also. Note your results in table 2.
7. Plot a graph between Absorbance on y-axis vs. Concentration on x-axis.

Observations :
Room Temp. -----^{0}C

Observation Table No. 1. Determination of λ_{max} for 0.0001 M $KMnO_4$

S. No.	Wavelength(nm)	Absorbance (A)

Observation Table No. 2. Absorbance of $KMnO_4$ solutions at λ_{max} (.............nm)

S. No.	Volume of 0.0001 M $KMnO_4$ (ml)	Volume of water (ml)	Concentration of $KMnO_4$ (M)	Absorbance (A)
1	10	0		
2	7.5	2.5		
3	5	5		
4	2.5	7.5		
5	1	9		
6	X	Y		

Result :
1. λ_{max} for 0.0001 M $KMnO_4$ solution = ...nm
2. Concentration of Unknown solution = ...M
3. Comment on the nature of the graph-

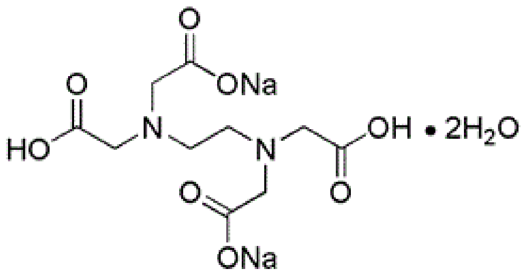
****

Group B
Exercise No. 1

Aim: Determination of total hardness by EDTA method.

Apparatus: Conical flask, pipette, burette, etc.
Solutions required: known water sample, unknown water sample, Eriochrome black – T indicator, EDTA solution, Buffer solution at pH 10.

Theory: Hardness of water can be determined by complex metric titration. Ethylene-diaminetetraacetic acid (EDTA) is a well known complexing agent, which is widely used in analytical work on account of its powerful complexing action and commercial availability.
The EDTA method is based upon the fact that ethylenediamine tetra-acetic acid (EDTA) forms remarkably stable complex compounds with all metals particularly with di and polyvalent metals.
Function of EDTA :- For all practical purposes the hydrated disodium salt of the reagent is used which has the following formula.



Chemical structure of disodium salt of EDTA

This formula may be abbreviated for the sake of simplicity as Na_2H_2Y and its dissociation at pH 10 is written as-

$$Na_2H_2Y \xrightarrow{pH\ 10} 2Na + H_2Y^{2-}$$

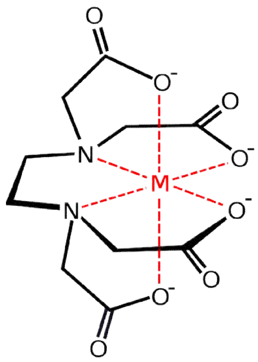
$$H_2Y^{2-} + Ca^{2+} \xrightarrow{pH\ 10} CaH_2Y\ (Ca - EDTA\ complex)$$

$$H_2Y^{2-} + Mg^{2+} \xrightarrow{pH\ 10} MgH_2Y\ (Mg - EDTA\ complex)$$

EDTA ion reacts with calcium and magnesium forming stable complexes, the calcium–EDTA complex is more stable then magnesium EDTA which is being more stable then magnesium indicator complex.

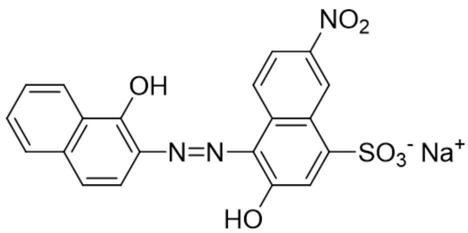
The relative stability of the complexes so far considered is as follows

$$Ca - EDTA\ \ complex > Mg - EDTA\ complex > Mg - Indicator\ complex$$



Chemical structure of Metal-EDTA complex

Function of indicator: The indicator Eriochrome black-T is a tribasic acid dye, the abbreviated formula of which is written as H_3In and which dissociates producing different colored ions at various pH as follows:



Chemical structure of eriochrome black-T indicator

$$H_3In \leftrightharpoons H_2In^- \leftrightharpoons HIn^{2-} \leftrightharpoons In^{3-}$$

H_2In^-	HIn^{2-}	In^{3-}
Red (below pH 6)	Blue (pH 7-11)	Orange (above pH 12)

The bivalent blue colored indicator ion which exists at about pH 10 reacts with magnesium producing red colored magnesium indicator complex:

HIn^{--} +	Mg^{++}	⟶	$MgIn^-$ +	H^+
Free Indicator (blue)	Analyte (colorless)		Complex (Wine red)	

So, if a drop of indicator is added to hard water sample containing Ca^{2+} & Mg^{2+} ions at pH 10, the magnesium indicator complex will be formed producing wine red color.
Now to this solution, if EDTA is gradually added, it will remove calcium ions first because Ca-EDTA complex is more stable. When all Ca^{2+} ions are used up, excess EDTA, if added will snatch Mg^{2+} ions from Mg indicator complex because Mg-EDTA complex is more stable than Mg-indicator complex leaving bivalent indicator ion free, which being blue in color will produce blue colored solution. Thus, a change from red to blue color will indicate the end point.

$$\underset{\text{Wine red}}{MgHIn} + H_2Y^{2-} \rightarrow MgH_2Y + \underset{\text{Blue}}{HIn^{2-}}$$

Procedure:

1. Pipette out 20 ml of the known water sample into a 100 ml conical flask.
2. Add 2 ml of the buffer solution and 2 drops of Eriochrome Black-T indicator. The solution becomes wine red.

3. Titrate it with EDTA solution from the burette until color changes to clear blue at the end point. Note the burette reading.
4. Repeat the same process, till you get two concordant readings.
5. Now titrate the unknown water sample in the similar manner till two concordant readings are obtained.

Observation for known solution:

S.No.	Volume of Known Solution (ml)	Volume of Intermediate Solution (ml)		Total Volume of Intermediate Solution (ml)
		Initial Reading	Final Reading	
1.				
2.				
3.				

Observation for Unknown solution:

S.No.	Volume of Known Solution (ml)	Volume of Intermediate Solution (ml)		Total Volume of Intermediate Solution (ml)
		Initial Reading	Final Reading	
1.				
2.				
3.				

Calculations:

$$N_1V_1 = N_2V_2$$
$$N_2 = \frac{N_1V_1}{V_2}$$

$$N_2V_2' = N_3V_3$$
$$N_3 = \frac{N_2V_2'}{V_3}$$

N_1 = Normality of known solution
V_1 = Volume of known solution
N_2 = Normality of intermediate solution
V_2 = Volume of intermediate solution
N_3 = Normality of alloy solution
V_3 = Volume of alloy solution
V_2' = Volume of intermediate solution with unknown solution

After knowing the normality of unknown solution (N_3), we can convert it into strength by multiplying it with equivalent weight of $CaCO_3$. Equivalent weight of $CaCO_3$ = 50.

N_3 x Equivalent weight of $CaCO_3$ = Strength in g/L of $CaCO_3$ Or Hardness of given water sample g/L.

N_3 x Eq. wt. of $CaCO_3$ x 10^3 = Hardness of given water sample in ppm or mg\L.

Result: The EDTA hardness of the given water sample is……… ppm.

****

Exercise No. 2

Aim: Determination of the carbonate, bicarbonate and total alkalinity of given water sample by neutralization titration.

Apparatus: Conical flask, Pipette, Burette, etc.
Solution required: N/30 Sodium carbonate solution, hydrochloric acid solution, Phenolphthalein and Methyl orange indicators.

Theory: The alkalinity of natural water is generally due to the presence of carbonates and bicarbonates of calcium and magnesium. In industries, hard water is softened by lime soda process due to which the resulting water contains considerable amount CO^{3-} and OH^{-} which lead to alkalinity. Highly alkaline water may lead to caustic embrittlement and also may cause deposition of precipitates and sludge in boiler tubes and pipes.
Depending on the anions that are present in water alkalinity is classified as:-
1. Hydroxides only
2. Carbonates only
3. Bicarbonates only
4. Hydroxides and Carbonates
5. Carbonates and bicarbonates

The possibility of hydroxides and bicarbonates existing together is ruled out because of the fact that they combine with each other as follows forming carbonates.

$$OH^- + HCO_3^- \rightarrow CO_3^{2-} + H_2O$$

The type and extent of alkalinity present in a water sample may be conveniently determined by titrating aliquot of the sample with a standard acid to phenolphthalein end-point, p and continuing the titration to the methyl orange end-point, m.

The reactions taking place may be represented by the following equations:

$$OH^- + H^+ \rightarrow H_2O \qquad (1)$$

$$CO_3^{2-} + H^+ \rightarrow HCO_3^- \qquad (2)$$

$$HCO_3^- H^+ \rightarrow H_2O + CO_2 \qquad (3)$$

The volume of acid run-down up to phenolphthalein end-point (p) corresponds to the complete neutralization of OH^- ions and half neutralization of CO^{3-} ions, while the volume of acid run after p corresponds to the complete neutralization of HCO_3^-. The total amount of acid used from beginning of the experiment corresponds to the total alkalinity present i.e. those formed from half neutralization of CO_3^{2-} and those already present in the water sample.

Procedure:
1. Pipette out 20 mL of the known solution into the conical flask.
2. Now add 2 drops of methyl orange indicator. Titrate the solution against acid solution till color changes from yellow to pink.
3. Note the volume of acid used.
4. Repeat the same experiment till two concordant readings are obtained.
5. Now pipette out 20 mL of unknown water sample into the conical flask. Add two drops of phenolphthalein indicator. The solution becomes pink indicating presence of CO_3^{2-} ions.
6. Titrate the pink solution against acid till solution becomes colorless. Note down the

reading of the burette.
7. At this stage add two drops of methyl orange indicator to the same solution. Titrate the solution against acid till solution becomes pink. Note the burette reading.
8. Repeat the experiment till two concordant readings are obtained.

Observation for known solution:

S.No.	Volume of Known Solution (ml)	Volume of Intermediate Solution (ml)		Total Volume of Intermediate Solution (ml)
		Initial Reading	Final Reading	
1.	20	0.0		
2.	20	0.0		
3.	20	0.0		

Observation for Unknown solution:

S.No.	Volume of unknown Solution (ml)	Volume of Intermediate Solution (ml)					
		Initial Reading	Final Reading	p	Initial Reading	Final Reading	m
1		0.0					
2		0.0					
3		0.0					

Calculation:

$$N_1V_1 = N_2V_2$$
$$N_2 = \frac{N_1V_1}{V_2}$$

N_1 = Normality of known solution
V_1 = Volume of known solution
N_2 = Normality of intermediate solution
V_2 = Volume of intermediate solution

Phenolphthalein Titration:

$$N_2 \times p = N_p \times V_3$$
$$N_p = \frac{N_2 \times p}{V_3}$$

N_2 = Normality of intermediate solution
p = Volume of intermediate solution with phenolphthalein indicator
N_p = Normality of unknown solution (p)
V_3 = Volume of unknown solution

Methyl Orange Titration:

$$N_2 \times m = N_m \times V_3$$
$$N_m = \frac{N_2 \times m}{V_3}$$

N_2 = Normality of intermediate solution
m = Volume of intermediate solution with methyl orange indicator
N_m = Normality of unknown solution (m)
V_3 = Volume of unknown solution

After knowing the normality of unknown solution (N_p, N_m), we can convert it into strength by multiplying it with equivalent weight of $CaCO_3$.

Phenolphthalein alkalinity, P (in ppm) = N_p x eq. weight of $CaCO_3$ x 10^3

Methyl orange alkalinity, M = N_m x eq. weight of $CaCO_3$ x 10^3

Result:

Total alkalinity (T)	=	P + M
Carbonate alkalinity	=	2P
Bicarbonate alkalinity	=	T – 2P

Exercise No. 3

Aim: Determination of the chloride present in given water sample by precipitation titration

Articles required:
Apparatus: Burette, Measuring cylinder, Beaker, Dropper and Stirrer
Chemicals: Potassium chromate indicator and Silver nitrate solution (0.0141 N)

Significance of determination: Chlorides in reasonable concentrations are not harmful to human. At concentrations above 250 mg/L, they can damage plants when used for gardening/irrigation and give a salty taste to water, which is objectionable to many people.

Theory/ Principle: (Mohr's Method)
This method determines the chloride ion concentration of a solution by titration with silver nitrate. As the silver nitrate solution is slowly added, a precipitate of silver chloride forms.

$Ag^{+}(aq) + Cl^{-}(aq) \rightarrow AgCl(s)$

The end point of the titration occurs when all the chloride ions are precipitated. Then additional silver ions react with the chromate ions of the indicator, potassium chromate, to form a red-brown precipitate of silver chromate.

$2Ag^{+}(aq) + CrO_4^{2-} \rightarrow Ag_2CrO_4(s)$

This method can be used to determine the chloride ion concentration of water samples from many sources such as seawater, stream water, river water and estuary water. The pH of the sample solutions should be between 6.5 and 10. If the solutions are acidic, the gravimetric method or Volhard's method should be used.
The end point of titration cannot be detected visually unless an indicator capable of demonstrating the presence of excess Ag+ is present. The indicator normally used is potassium chromate, which supplies chromate ions. As the concentration of Cl^- ions becomes exhausted, the silver ion concentration increases and a reddish brown precipitate of silver chromate is formed.

$2Ag^{+} + CrO_4^{2-} \rightarrow Ag_2CrO_4$ **(reddish brown precipitate)**

This is taken as evidence that all chloride has been precipitated. Since an excess Ag^+ is needed to produce a visible amount of Ag_2CrO_4, the indicator error is subtracted from all titrations. The indicator error or blank varies somewhat with the ability of individuals to detect a noticeable color change. The usual range is 0.2 to 0.4 mL of titrant.

Procedure:
1. Take 50 mL of the sample in a beaker and add 5 drops (about 1 mL) of potassium chromate indicator to it.
2. Add standard (0.0141 N) silver nitrate solution to the sample from a burette, a few drops at a time, with constant stirring until the first permanent reddish color appears. This can be determined by comparison with distilled water blank. Record the volume of silver nitrate used.
3. If more than 7 or 8 mL of silver nitrate solution is required, the entire procedure should be repeated using a smaller sample diluted to 50 ml with distilled water.

Observation for Unknown solution:

S.No.	Volume of Known Solution (ml)	Volume of silver nitrate solution(ml)		Total Volume of Silver nitrate solution (ml)
		Initial Reading	Final Reading	
1.				
2.				
3.				

Calculations:
Chloride, Cl^- (mg/L)

= (mL of $AgNO_3$used - "error" or "blank") x Multiplying Factor (M.F.)

Where,

$$\textbf{M.F.} = \frac{\text{Normality of } AgNO_3 \text{ x equivalent wt.of } Cl^- \text{ x } 1000}{\text{mL of sample taken}}$$

Result: The concentration of chloride ion in the the given water sample is……… mg/L.

Exercise No.4

Aim: To determine percentage purity of Iron alloy by internal indicator method.

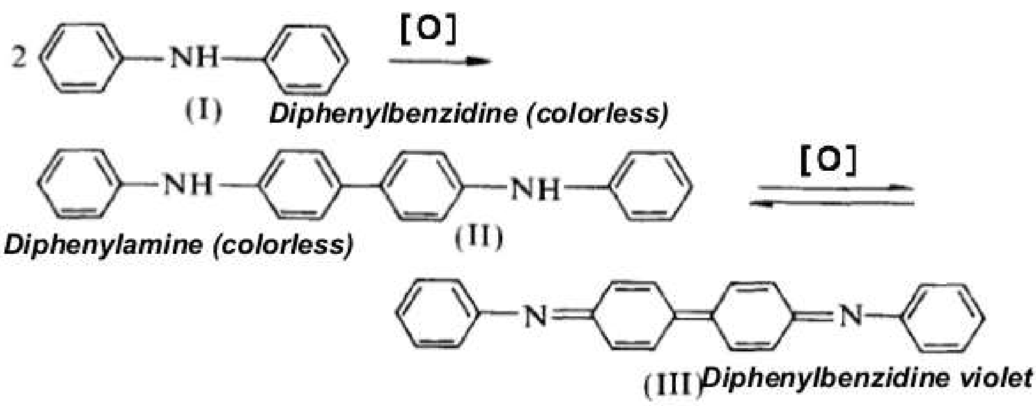
Apparatus: Conical flask, Pipette, Burette, etc.
Solutions required: N/40 ferrous ammonium sulphate solution, diphenylamine as an internal indicator, potassium dichromate solution, a mixture of H_2SO_4 & H_3PO_4, alloy solution (concentration 4 g/L).

Theory:
$K_2Cr_2O_7$ contains oxygen available for oxidation which it loses in presence of a substance capable for receiving it, and in presence of an excess of sulphuric or hydrochloric acid. Thus a ferrous salt in an acidified solution is readily oxidized to ferric salt by the addition of a solution of potassium dichromate.

$$FeSO_4\,(NH_4)_2SO_4.6H2O \rightarrow FeSO_4 + (NH_4)_2SO_4 + 6H_2O$$
$$6FeSO_4 + K_2Cr_2O_7 + 7H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + K_2SO_4 + 7H_2O$$

At the beginning of titration there are only ferrous ions, as the titration proceeds these ions are converted into ferric ions. At the end point all the ferrous ions are converted into ferric ions. Gram equivalent weight of the $K_2Cr_2O_7$ used upto the end point will correspond to the gram equivalent of ferrous ions present in the solution of alloy.

The end point of the reaction can be detected using diphenylamine as an indicator, 1% solution of diphenylamine in concentrated sulphuric acid is used for the titration of iron (II) with potassium dichromate solution. Diphenylamine is first as oxidized in to biphenyl benzidine (colorless) which is the real indicator. It is reversibly oxidized to its quinonoid form (violet color). As it shows different colors in reduced & oxidized form, it acts as a redox indicator.



Red-Ox potential of **In (0.5-1M H_2SO_4 + 0.5M H_3PO_4)** medium,
$E^{o/}_{In}$ = - 0.76 V

Suitable for titration of Fe^{+2} by dichromate soln, in presence of H_3PO_4
$E^{o/}_{In}$ = - 0.61 V

During the titration, oxidation of two species takes place. The first one is the oxidation of ferrous into ferric ions and another one is the oxidation of diphenylamine into diphenylbenzidine.
The oxidation potential of the two systems is given by the Nernst's equation:

$$E = E^o + \frac{RT}{nf}\ln\frac{[Oxid.]}{[Red.]}$$

And the formal potential of system:
$Fe^{2+} \rightarrow Fe^{3+}$ is 0.68 V.

Similarly, formal potential of indicator system is 0.76 V. The system having low oxidation potential will oxidize first. Due to oxidation of Fe^{2+} into Fe^{3+}, concentration of Fe^{3+} increases. It results in the increase in the potential of the system. If the potential of the system exceeds 0.76 V, it results in the oxidation of indicator even before the actual end point. So phosphoric acid is added to reduce the formal potential of ferrous-ferric ion system.
Excess of Fe^{3+} ions are removed by phosphoric acid which forms stable green colored complex with Fe^{3+}:

$$H_3PO_4 + Fe^{3+} \rightarrow \underbrace{(FeHPO_4)^+}_{\text{Green Colored Complex}} + 2H^+$$

Due to the formation of this complex the potential of the system remain below 0.76 V till the end point. At end point, one drop of $K_2Cr_2O_7$ will oxidize the indicator and color of the solution changes to blue violet.

Procedure:
1. Pipette out 20 ml of the known solution into the conical flask. Add about 2 ml of acid mixture and 2 drops of diphenylamine indicator into it.
2. Now titrate it against $K_2Cr_2O_7$ solution till solution becomes blue. Note down the reading and repeat the experiment for two concordant observations.
3. Similarly, repeat the experiment with unknown alloy solution.

Observation for known solution:

S.No.	Volume of Known Solution (ml)	Volume of $K_2Cr_2O_7$ Solution (ml)		Total Volume of $K_2Cr_2O_7$ Solution used (ml)
		Initial Reading	Final Reading	
1.	20	0.0		
2.	20			
3.	20			

Observation for unknown solution:

S.No.	Volume of Known Solution (ml)	Volume of $K_2Cr_2O_7$ Solution (ml)		Total Volume of $K_2Cr_2O_7$ Solution used (ml)
		Initial Reading	Final Reading	
1.	20	0.0		
2.	20			
3.	20			

Calculations:

$$N_1V_1 = N_2V_2$$
$$N_2 = \frac{N_1V_1}{V_2}$$

$$N_2V_2' = N_3V_3$$
$$N_3 = \frac{N_2V_2'}{V_3}$$

N_1 = Normality of known solution
V_1 = Volume of known solution
N_2 = Normality of intermediate solution
V_2 = Volume of intermediate solution
N_3 = Normality of alloy solution
V_3 = Volume of alloy solution
V_2' = Volume of intermediate solution with alloy solution

Strength in g/L = Normality x equivalent weight of Iron

% of Iron in given alloy solution = Strength in g/L x 55.8 x 100/4

Result: The percentage of iron in the given alloy sample is……….

Exercise No.5

Aim: To determine percentage purity of iron alloy by external indicator method.

Apparatus: Conical flask, Pipette, Burette, etc.
Solutions required: N/40 ferrous ammonium sulphate solution, potassium ferricyanide as an external indicator, potassium dichromate solution, dilute H_2SO_4 solution, alloy solution (concentration 4 g/L).

Theory: $K_2Cr_2O_7$ contains oxygen available for oxidation which it loses in presence of a substance capable for receiving it, and in presence of an excess of sulphuric or hydrochloric acid. Thus a ferrous salt in an acidified solution is readily oxidized to ferric salt by the addition of a solution of potassium dichromate.

$$FeSO_4\,(NH_4)_2SO_4.6H2O \rightarrow FeSO_4 + (NH_4)_2SO_4 + 6H_2O$$
$$6FeSO_4 + K_2Cr_2O_7 + 7H_2SO_4 \rightarrow 3Fe_2(SO_4)_3 + Cr_2(SO_4)_3 + K_2SO_4 + 7H_2O$$

At the beginning of titration there are only ferrous ions, as the titration proceeds these ions are converted into ferric ions. At the end point all the ferrous ions are converted into ferric ions. Gram equivalent weight of the $K_2Cr_2O_7$ used up to the end point will correspond to the gram equivalent of ferrous ions present in the solution of alloy.
Conversion of whole of the ferrous ion to ferric ion is determined by using potassium ferricyanide as an external indicator. Ferrous ions but not ferric ions give a blue precipitate with this indicator. The dichromate solution is accordingly added to acidified iron solution, until a drop of the liquid ceases to give a blue coloration when it is brought into the contact of a drop of potassium ferricyanide solution placed on a glazed white tile.
The formation of blue precipitates occurs due to the following reaction:

$$K_3[Fe^{3+}(CN)_6] + Fe^{2+}SO_4 \rightarrow \underset{\substack{\text{Potassium ferro-ferricyanide} \\ \text{(Prussian blue)}}}{KFe^{2+}[Fe^{3+}(CN)_6]} + K_2SO_4$$

Procedure:
1. Five separate drops of the potassium ferricyanide indicator are placed on a clean and dry white glazed tile in an orderly row.
2. Pipette out 20 ml of known solution into the conical flask and acidify it by adding dilute acid (about half test tube).
3. Fill the burette with intermediate $K_2Cr_2O_7$ solution with the help of funnel. Remove funnel from top and the air bubble, if any, from the nozzle of the burette.
4. Now add 5 ml of the dichromate solution in the acidified known solution from a burette with constant stirring.
5. Withdraw a drop of this solution with a glass rod and mix it with one of the drops of indicator solution on the tile. If a blue color or precipitate is seen, an additional 5 ml of dichromate solution should be added from the burette and tested again, and so on, adding the dichromate solution in volumes of 5 ml at a time until only a greenish yellow color is seen on the tile.
In this way the end point is 'bracketed' first within the limit of 5 ml . Suppose the end point is found to be between 15 ml and 20 ml.
6. Now, fresh 20 ml of known solution is taken, acidified and then 15 ml of $K_2Cr_2O_7$

solution is added directly without testing. The end point is tested by adding 1 ml of intermediate solution, every time and so on, until a new ‘bracket’ of one ml is found. Let it be 16-17 ml.
7. A third volume of known solution is taken, acidified and 16 ml of dichromate solution is added to conical flask directly. The end point is tested by adding 0.1 ml of intermediate at a time until the disappearance of the blue color on the tile is obtained. The last result is checked by the fourth volume of the known solution.
8. The unknown alloy solution is now, titrated with a standard dichromate solution by repeating the above steps.

Observation for known solution:

S.No.	Volume of Known Solution (ml)	Volume of $K_2Cr_2O_7$ Solution (ml)		Total Volume of $K_2Cr_2O_7$ Solution used (ml)	Color observed
		Initial Reading	Final Reading		
1 2 3 4	20	0.0	5.0 10.0 15.0 20.0	5.0 10.0 15.0 20.0	Blue Blue Blue Greenish yellow
1 2 3	20	0.0	15.0 16.0 17.0	15.0 16.0 17.0	Blue Blue Greenish yellow
1 2 3	20	0.0	16.1 16.2 16.3	16.1 16.2 16.3	Blue Blue Greenish yellow
1 2 3	20	0.0	16.1 16.2 16.3	16.1 16.2 16.3	Blue Blue Greenish yellow

Observation for unknown solution:

S.No.	Volume of Known Solution (ml)	Volume of $K_2Cr_2O_7$ Solution (ml)		Total Volume of $K_2Cr_2O_7$ Solution used (ml)	Color observed
		Initial Reading	Final Reading		
1 2 3	20	0.0	5.0 10.0 15.0	5.0 10.0 15.0	Blue Blue Greenish yellow
1 2 3	20	0.0	10.0 11.0 12.0	10.0 11.0 12.0	Blue Blue Greenish yellow
1 2	20	0.0	11.1 11.2	11.1 11.2	Blue Greenish yellow
1 2	20	0.0	11.1 11.2	11.1 11.2	Blue Greenish

					yellow

Calculations:

$$N_1V_1 = N_2V_2$$
$$N_2 = \frac{N_1V_1}{V_2}$$

$$N_2V_2' = N_3V_3$$
$$N_3 = \frac{N_2V_2'}{V_3}$$

N_1 = Normality of known solution
V_1 = Volume of known solution
N_2 = Normality of intermediate solution
V_2 = Volume of intermediate solution
N_3 = Normality of alloy solution
V_3 = Volume of alloy solution
V_2' = Volume of intermediate solution with alloy solution

Strength in g/L = Normality x equivalent weight of Iron

% of Iron in given alloy solution = Strength in g/L x 55.8 x 100/4

Result: The percentage of iron in the given alloy sample is……….

Exercise No. 12

Aim: To study the Chemical oscillations of Iodine clock reaction.

Articles required:

(a) **Apparatus:** Beakers, Graduated cylinder, and Stir plate or stir bar

(b) **Chemicals:** H_2O_2, KIO_3, H_2SO_4, $MnSO_4 \cdot H_2O$, Malonic acid, Soluble starch, and Deionized water.

Significance: Study of oscillatory reaction completed. Demonstration of two reactions which can switch back and forth in which the product of one reaction will be the reactant for the other.

Theory/ Principle:

Briggs-Rauscher Reaction –It is Oscillation time (switch time) in which solution turns from yellow to blue color as per following reactions.

$$IO_3^- + 2\,H_2O_2 + CH_2(CO_2H)_2 + H^+ = ICH(CO_2H)_2 + 2\,O_2 + 3\,H_2O$$

The BR reaction can be summarized by two chemical reactions:

$$IO_3^- + 2\,H_2O_2 + H^+ = HOI + 2\,O_2 + 2\,H_2O \quad \text{----->} \quad (A)$$

$$HOI + CH_2(CO_2H)_2 = ICH(CO_2H)_2 + H_2O \quad \text{----->} \quad (B)$$

When the reaction begins, the solution becomes yellow a $I^{-($ generated from reaction of HOI with H_2O_2, not shown) reacts with HOI under acidic conditions to produce I_2. The solution suddenly changes to dark blue as a buildup of I^- reacts with I_2 in the presence of starch to generate the pentaiodide ion (I_5^-) which is encased in the amylose present in solution. The reaction proceeds by consuming I_2 at a rate greater than it is being produced thus removing the presence of $I_5^-$and increasing the concentration of I^-(colorless). The oscillations continue until malonic acid or $IO_3^-$is completely consumed.

Solution Prepration

1. Solution A: Prepare 100 mL by diluting 30 mL of 30% H_2O_2 with 70 mL of deionized H_2O.
2. Solution B: Prepare solution by adding 10 mL of 1.0 M H_2SO_4 to 80 mL of deionized water. Dissolve 4.3 g KIO_3 in this solution and dilute to 100 mL.
3. Solution C: Prepare starch solution by dissolving 0.1 g of soluble starch in 90 mL of boiling deionized water. When cool, add 1.5 g malonic acid, 0.4 g $MnSO_4 \cdot H_2O$, stir and dilute to 100 mL.

Procedure:

Add 50 mL of 9.0 % H_2O_2 Solution to a clean beaker fitted with a stir bar. Then add 50 mL of an acidified 0.2 M KIO_3 Solution, mix thoroughly then add 50 mL of 0.9 % Starch and stir it. Upon addition of the final solution, bubbles should appear. Initially solution turns yellow then blue and finally colorless. This reaction will oscillate for 5-10 minutes.Take atleast 10 observations

Observations:
Room Temp. -----°C

Observation Table-

S. No.	Oscillation time (minutes) (Yellow to Blue color switch time)	Average Oscillation Time(minutes)
1		
2		
3		

Oscillation time (Yellow to Blue color switch time) ---------------minutes

Result:

The three colors observed are a result of an increase in the concentrations of I_2 (yellow), starch-iodine complex (blue), and I^- (colorless) in order of appearance during the demonstrat and demonstrated in Lab.

Precautions:

1. Wear proper protective equipment including gloves and safety glasses when preparing and performing this demonstration. Concentrated hydrogen peroxide can cause burns. KIO_3 is an oxidizer. Sulfuric acid is severely, corrosive to eyes, skin and other tissue. Malonic acid solution is moderately toxic and corrosive to eyes, skin and respiratory tract. The reaction produces iodine which is toxic by inhalation and irritating to eyes, skin, and respiratory tract.
2. Perform the demonstration in a well-ventilated room.
3. Disposal of final solution should be in an appropriate aqueous waste container.

Exercise No. 13

Aim: To determine the amount of ferrous ammonium sulphate (FAS) present in the given solution using standard $K_2Cr_2O_7$

Articles required:

Apparatus: potentiometer (pH meter) *measuring cylinder. Beakers, Burette.etc*

Chemicals: Ferrous Ammonium Sulphate, Potassium Dichromate, H_2SO_4, Deionized water.

Principle: The titrations involving the determination of end point of redox system by the measurement of e.m.f. of the cell are called potentiometric titrations. In the case of oxidation-reduction titration of FAS using standard dichromate solution, a standard calomel electrode (SCE) is used as the reference electrode, whose E^0 remains constant.

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

From Nernst's equation we have:

$$E = E^o + \frac{RT}{nf}\ln\frac{[Fe^{3+}]}{[Fe^{2+}]}$$

As the oxidation of Fe^{2+} progress, ΔE value increases slowly in the beginning and then sharply changes near the equivalence point.

Procedure:

1. Pipette out exactly 25 mL of given FAS solution in a clean 100 mL beaker and add 1 test tube full of dil. H_2SO_4 to maintain acidic media.

2. Rinse the standard calomel electrode and bright platinum electrode assembly of pH meter with ion exchange water and immerse them in the reaction mixture. Connect the electrode assembly to the (in mV mode). Switch on the potentiometer and record the potential.

3. Add standard $K_2Cr_2O_7$ from the burette in 0.5mL quantity at a time and stir the solution using a glass rod for uniform mixing. Record the potential value (E) after each addition of $K_2Cr_2O_7$.

4. Continue the procedure till the potential increases more rapidly near the equivalence point. Once equivalence is past there will be only a slight increase in the potential for the successive addition of $K_2Cr_2O_7$ solution.

5. Plot a graph of ΔE/ ΔV values (along y-axis) versus volume of $K_2Cr_2O_7$ (along x-axis). From the graph find out the volume of $K_2Cr_2O_7$ solution required for complete oxidation (peak value of sharp curve). Finally, calculate the normality of FAS and amount of ion present in the given solution.

Calculations: Calculate the amount of present in the given solution using following formula

$$N_2 = \frac{N_1V_1}{V_2}$$

Where,

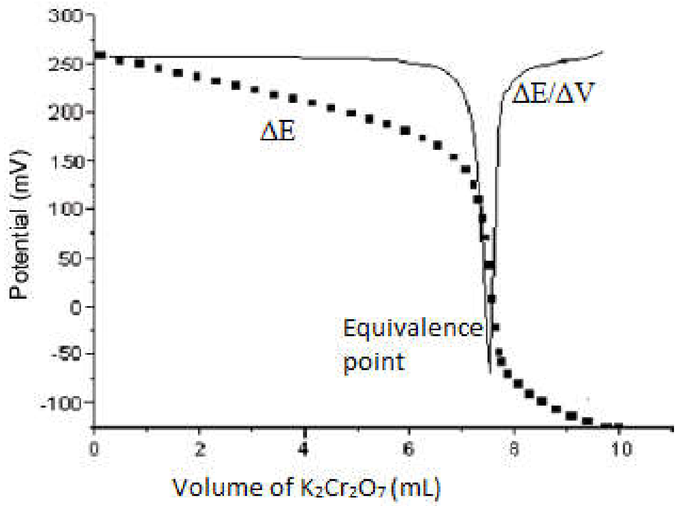
N_1 = Normality of $K_2Cr_2O_7$ solution (given)
V_1 = Volume of $K_2Cr_2O_7$ used
(cm^3 of $K_2Cr_2O_7$ solution corresponding to the equivalence point from graph)

V_2 = Volume of FAS solution
N_2 = Normality of FAS solution

Then

Weight of ferrous ammonium sulphate present in one liter of its solution
= N_2 x equivalent weight of FAS
= N_2 x 392.14 g/L

Volume of $K_2Cr_2O_7$	EMF in mV	ΔE	$\Delta E/\Delta V$
0.0			
0.5			
1.0			
1.5			



Result:

1. The volume of $K_2Cr_2O_7$ solution corresponding to the equivalence point = ……………….. mL
2) Normality of given FAS solution (N_2) = ………………. mL
3) Amount of FAS present in litre =…………… g/L

Precautions

Exercise No. 14

Aim: (i) To determine the distribution coefficient of iodine between CCl_4 and water at room temperature.
(ii) To find the equilibrium constant of above distribution.
(iii) To find the concentration of given KI solution.

Articles required:

Apparatus: Stoppered bottles, mechanical shaker**,** separating funnel, conical flask, beaker and titration tiles etc.
Chemicals: Iodine, KI, CCl_4, sodium thiosulphate solution (hypo)**, s**tarch and distilled water**,**

Theory/ Principle: According to **Nernst distribution law**, when a solute distributes between two immiscible solvents in contact with each other, there exists, for similar molecular species, at a given temperature, a constant ration of distribution between the two solvents irrespective of the total amount of the solute and any other molecular species which may be present. In other words,

$$C_1/C_2 = K_D$$

Where K_D is termed as the distribution coefficient, and, the terms C_1 and C_2 refer to the concentrations of similar molecular species in the two liquids at a constant temperature.
Here we are considering a system of distribution of iodine between aqueous and CCl_4 layer in the presence of KI in aqueous layer. Equilibrium between KI and I_2 will be attained in the aqueous layer. The equilibrium attained is

$$\mathbf{KI + I_2 \rightarrow KI_3^-}$$

The equilibrium constant

$$\mathbf{K_{equ.} = [KI_3^-]/ [KI]\cdot[I_2]}$$

Procedure:

1. Serially labelled three bottles are taken and mixtures are made in each bottle as follows:

Bottle No.	Contents
I	20 ml of I_2 in CCl_4 + 40 ml of water
II	20 ml of I_2 in CCl_4 + 40 ml of known KI
III	20 ml of I_2 in CCl_4 + 40 ml of unknown KI

2. These bottles are stoppered well and shaken in a mechanical shaker for about an hour.

After shaking, they are set aside for about 20 minutes in a water trough so that it will attain equilibrium.

3. Exactly 2 ml of the CCl_4 layer from bottle I is pipetted out into a conical flask and approximately 20 ml of distilled water is added this biphasic solution is titrated against previously standardized thio solution using starch as indicator. The end point is the disappearance of blue color.

4. The experiments are repeated to get concordant values.

5. From bottle I, about 10 ml of the aqueous layer is pipetted out and the amount of I_2 is estimated as done with CCl_4 layer. The above procedure is repeated for bottles II and III.

Observations:
Room Temp. -----°C
Observation Table No. (1)

Bottle No.	**Volume of CCl_4 layer(ml)**	**Volume of Thio used by Burette**			**Concordant Value (ml)**
		Initial	**Final**	**Difference**	
I	2.0				
	2.0				
II	2.0				
	2.0				
III	2.0				
	2.0				

Bottle No.	**Volume of water layer(ml)**	**Volume of Thio used by Burette**			**Concordant Value (ml)**
		Initial	**Final**	**Difference**	
I	**10.0**				
	10.0				
II	**10.0**				
	10.0				
III	**10.0**				
	10.0				

Observation Table No. (2)

Bottle No	**Layer**	**Volume of Thio (ml)**	**Strength of Iodine**	**$K_D = C_1/C_2$**
I	CCl_4			
	Aqueous			

II	CCl_4			
	Aqueous			
III	CCl_4			
	Aqueous			

Result:

(i) The distribution coefficient of iodine between CCl_4 and water is found to be __________

(ii) The equilibrium constant, K_{eq}, for above equilibrium is found to be ____M^{-1}.

(iii) The concentration of given KI solution is found to be _________ M.

Precautions: Wear proper protective equipment including gloves and safety glasses when preparing and performing this demonstration. Iodine is corrosive to eyes, skin and other tissue. The reaction uses CCl_4which is toxic by inhalation.

Group A

1. Determination of viscosity of lubrication oil by Redwood viscometer no. 1 (At five different temperatures).
2. Determination of percentage of moisture in a coal sample
3. Determination of flash point of a given oil by Abel's apparatus
4. Determination of flash point of a given oil by Pensky Martin's apparatus
5. Determination of total solids in a water sample
6. Determination of aniline point of a given oil sample
7. Determination of Drop point of a semi-solid lubricant
8. Determination of acid value of an oil sample
9. Determination of viscosity of lubrication oil by Redwood viscometer no. 2. (at five different temperatures)
10. Determination of Steam emulsification number (SEN) of a given lubricating oil sample
11. To study the Chemical oscillations (Iodine clock reaction)
12. Potentiometric estimation of Ferrous Ammonium Sulphate using standard Potassium Dichromate Solution
13. Determination of the partition coefficient of a substance between two immiscible liquids.
14. Synthesis of polymer and nanomaterials
15. Verification of Beer- Lambert's law by visible spectroscopy

Group B

1 Determination of hardness of water by EDTA method

2 Determination of carbonates, bicarbonates and total alkalinity of a water sample

3 Determination of chloride content of water

4 Determination of percentage purity of iron alloy by internal indicator method

5 Determination of percentage purity of iron alloy by external indicator method

Course Outcomes: This course will illustrate the principles of chemistry relevant to the study of science and engineering.The students will learn to Properties,prectical applications, analysis and determination of engineering materials like

- Lubricants (liquids and semisolid)
- Fuel oil & Coal samples,
- Water in terms of hardness, chloride content and alkalinity
- Polymers
- Application of spectroscopic techniques for